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Effect of door-to-balloon time on mortality in patients with ST-segment elevation myocardial infarction

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Abstract

Objectives: We sought to determine the effect of door-to-balloon time on mortality for patients with ST-segment elevation myocardial infarction (STEMI) undergoing primary percutaneous coronary intervention (PCI).

Background: Studies have found conflicting results regarding this relationship.

Methods: We conducted a cohort study of 29,222 STEMI patients treated with PCI within 6 h of presentation at 395 hospitals that participated in the National Registry of Myocardial Infarction (NRMI)-3 and -4 from 1999 to 2002. We used hierarchical models to evaluate the effect of door-to-balloon time on in-hospital mortality adjusted for patient characteristics in the entire cohort and in different subgroups of patients based on symptom onset-to-door time and baseline risk status.

Results: Longer door-to-balloon time was associated with increased in-hospital mortality (mortality rate of 3.0%, 4.2%, 5.7%, and 7.4% for door-to-balloon times of < or =90 min, 91 to 120 min, 121 to 150 min, and >150 min, respectively; p for trend <0.01). Adjusted for patient characteristics, patients with door-to-balloon time >90 min had increased mortality (odds ratio 1.42; 95% confidence interval [CI] 1.24 to 1.62) compared with those who had door-to-balloon time < or =90 min. In subgroup analyses, increasing mortality with increasing door-to-balloon time was seen regardless of symptom onset-to-door time (< or =1 h, >1 to 2 h, >2 h) and regardless of the presence or absence of high-risk factors.

Conclusions: Time to primary PCI is strongly associated with mortality risk and is important regardless of time from symptom onset to presentation and regardless of baseline risk of mortality. Efforts to shorten door-to-balloon time should apply to all patients.

Comment in

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